Variance of the Centered Oracle LDSC Estimator

1 Setup and Notation

Let $\mathbf{X} \in \mathbb{R}^{n\times p}$ be the genotype matrix whose rows $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are independent and identically distributed as

$$\mathbf{x}_i \sim N(\mathbf{0},\, \boldsymbol{\Sigma}), \qquad i = 1, \ldots, n,$$

where $\boldsymbol{\Sigma} \in \mathbb{R}^{p\times p}$ is a known, positive-definite, block-diagonal covariance matrix. The phenotype vector is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\alpha} + \boldsymbol{\epsilon},$$

where $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_p)^\top$ is a *fixed* (non-random) vector of genetic effects and $\boldsymbol{\epsilon} = (\epsilon_1, \ldots, \epsilon_n)^\top$ consists of independent $N(0, \sigma_\epsilon^2)$ errors, independent of $\mathbf{X}$.

Key derived quantities.

Definition 1 (Marginal signal and noise)**.** Define

$$\beta_j \;=\; (\boldsymbol{\Sigma}\boldsymbol{\alpha})_j \;=\; \sum_{k=1}^{p} \Sigma_{jk}\, \alpha_k, \qquad j = 1, \ldots, p,$$

and write $\boldsymbol{\beta} = \boldsymbol{\Sigma}\boldsymbol{\alpha} \in \mathbb{R}^p$. Define also the marginal phenotypic variance

$$\sigma_y^2 \;=\; \boldsymbol{\alpha}^\top \boldsymbol{\Sigma}\, \boldsymbol{\alpha} + \sigma_\epsilon^2.$$

Definition 2 (LD scores and estimator)**.** For $j = 1, \ldots, p$,

$$\ell_j = \sum_{i=1}^{p} \Sigma_{ji}^2, \qquad\qquad \widehat{a}_j = n^{-1}\mathbf{X}_j^\top \mathbf{y}, \qquad\qquad \widehat{w}_j = \widehat{a}_j^2, \tag{1}$$

where $\mathbf{X}_j$ denotes the j-th column of $\mathbf{X}$. Set $\boldsymbol{\ell} = (\ell_1, \ldots, \ell_p)^\top$, $\widehat{\mathbf{w}} = (\widehat{w}_1, \ldots, \widehat{w}_p)^\top$, and

$$\mu_\ell = \frac{1}{p}\, \mathbf{1}_p^\top \boldsymbol{\ell}, \qquad \widetilde{\boldsymbol{\ell}} = \boldsymbol{\ell} - \mu_\ell\, \mathbf{1}_p.$$

The centered oracle LDSC estimator is

$$\widetilde{\sigma}_\alpha^2 = \frac{\widetilde{\boldsymbol{\ell}}^\top (\widehat{\mathbf{w}} - \widehat{\mu}_w\, \mathbf{1}_p)}{\widetilde{\boldsymbol{\ell}}^\top \widetilde{\boldsymbol{\ell}}}. \tag{2}$$

Remark 3. Since $\widetilde{\boldsymbol{\ell}}^\top \mathbf{1}_p = \mathbf{1}_p^\top \boldsymbol{\ell} - p\,\mu_\ell = 0$, the intercept–centering term $\widehat{\mu}_w\, \mathbf{1}_p$ drops out:

$$\widetilde{\sigma}_\alpha^2 = \frac{\widetilde{\boldsymbol{\ell}}^\top \widehat{\mathbf{w}}}{\|\widetilde{\boldsymbol{\ell}}\|^2} = \frac{\displaystyle\sum_{j=1}^{p} \widetilde{\ell}_j\, \widehat{a}_j^2}{\displaystyle\sum_{j=1}^{p} \widetilde{\ell}_j^2}. \tag{3}$$

Because $\boldsymbol{\ell}$ depends only on the population covariance $\boldsymbol{\Sigma}$, the denominator $\|\widetilde{\boldsymbol{\ell}}\|^2$ is deterministic.

2 Preliminary Moment Calculations

Throughout, we write $y_i = \mathbf{x}_i^\top \boldsymbol{\alpha} + \epsilon_i$ and note that $(\mathbf{x}_i, \epsilon_i)$, $i = 1, \dots, n$, are mutually independent.

2.1 Representation as an i.i.d. average

Define, for each sample i and variant j,

$$T_{ij} = X_{ij}\, y_i = X_{ij}\big(\mathbf{x}_i^\top \boldsymbol{\alpha} + \epsilon_i\big). \tag{4}$$

Then $\widehat{a}_j = n^{-1} \sum_{i=1}^n T_{ij}$, and for each fixed j the random variables $T_{1j}, \dots, T_{nj}$ are i.i.d. copies of

$$T_j = X_j\, (\mathbf{x}^\top \boldsymbol{\alpha} + \epsilon), \tag{5}$$

where $\mathbf{x} \sim N(0, \boldsymbol{\Sigma})$ and $\epsilon \sim N(0, \sigma_\epsilon^2)$ are independent.

2.2 First and second moments of T_j

Lemma 4. *For all $j, k \in \{1, \dots, p\}$,*

$$\mathbb{E}[T_j] = \beta_j, \tag{6}$$

$$\gamma_{jk} \;:=\; \mathrm{Cov}(T_j, T_k) = \Sigma_{jk}\, \sigma_y^2 + \beta_j\, \beta_k. \tag{7}$$

Proof. Since ϵ is independent of $\mathbf{x}$ with mean zero,

$$\mathbb{E}[T_j] = \mathbb{E}[X_j\, \mathbf{x}^\top \boldsymbol{\alpha}] = \sum_m \alpha_m\, \mathbb{E}[X_j X_m] = \sum_m \alpha_m\, \Sigma_{jm} = \beta_j.$$

For the second moment,

$$\mathbb{E}[T_j\, T_k] = \mathbb{E}\big[X_j\, X_k\, (\mathbf{x}^\top \boldsymbol{\alpha} + \epsilon)^2\big] = \mathbb{E}\big[X_j X_k\, (\mathbf{x}^\top \boldsymbol{\alpha})^2\big] + \sigma_\epsilon^2\, \mathbb{E}[X_j X_k].$$

By Isserlis' theorem (fourth moment of Gaussians),

$$\begin{aligned}
\mathbb{E}\big[X_j X_k\, (\mathbf{x}^\top \boldsymbol{\alpha})^2\big] &= \sum_{m,l} \alpha_m \alpha_l\, \mathbb{E}[X_j X_k X_m X_l] \\
&= \sum_{m,l} \alpha_m \alpha_l \big(\Sigma_{jk}\Sigma_{ml} + \Sigma_{jm}\Sigma_{kl} + \Sigma_{jl}\Sigma_{km}\big) \\
&= \Sigma_{jk}\, (\boldsymbol{\alpha}^\top \boldsymbol{\Sigma} \boldsymbol{\alpha}) + 2\, \beta_j\, \beta_k.
\end{aligned}$$

Hence $\mathbb{E}[T_j T_k] = \Sigma_{jk}(\boldsymbol{\alpha}^\top \boldsymbol{\Sigma} \boldsymbol{\alpha} + \sigma_\epsilon^2) + 2\beta_j\beta_k = \Sigma_{jk}\sigma_y^2 + 2\beta_j\beta_k$, and $\gamma_{jk} = \mathbb{E}[T_j T_k] - \beta_j\beta_k = \Sigma_{jk}\sigma_y^2 + \beta_j\beta_k$. □

2.3 Third cross-central moments of T_j

Lemma 5. *Define $U_j = T_j - \beta_j$. For all $j, k \in \{1, \dots, p\}$,*

$$\rho_{jkk} \;:=\; \mathbb{E}[U_j\, U_k^2] = 4\, \sigma_y^2\, \Sigma_{jk}\, \beta_k + 2\, \sigma_y^2\, \Sigma_{kk}\, \beta_j + 2\, \beta_j\, \beta_k^2. \tag{8}$$

Proof. Write $W_j = T_j = X_j\, y$. Then $\rho_{jkk} = \mathbb{E}[W_j W_k^2] - 2\beta_k\, \mathbb{E}[W_j W_k] + \beta_k^2\, \beta_j - \beta_j\, \mathbb{E}[W_k^2] + 2\beta_j \beta_k^2 - \beta_j \beta_k^2$. Simplifying,

$$\rho_{jkk} = \mathbb{E}[W_j W_k^2] - 2\beta_k\, \gamma_{jk} - \beta_j\, \gamma_{kk} - \beta_j \beta_k^2. \tag{9}$$

Since $W_j W_k^2 = X_j X_k^2\, y^3$ and $y^3 = (\mathbf{x}^\top \boldsymbol{\alpha} + \epsilon)^3$, we expand using independence of ϵ and the fact that $\mathbb{E}[\epsilon] = \mathbb{E}[\epsilon^3] = 0$:

$$\mathbb{E}[W_j W_k^2] = \mathbb{E}\big[X_j X_k^2\, (\mathbf{x}^\top \boldsymbol{\alpha})^3\big] + 3\, \sigma_\epsilon^2\, \mathbb{E}[X_j X_k^2\, \mathbf{x}^\top \boldsymbol{\alpha}].$$

Sixth–moment term. Setting $s = \mathbf{x}^\top \boldsymbol{\alpha}$, Isserlis' theorem applied to $\mathbb{E}[X_j X_k X_k X_{m_1} X_{m_2} X_{m_3}]$ (summed against $\alpha_{m_1} \alpha_{m_2} \alpha_{m_3}$ over all 15 complete pairings of six Gaussian factors) yields, after collecting terms,

$$\mathbb{E}[X_j X_k^2\, s^3] = 6\, \Sigma_{jk}\, \beta_k\, (\boldsymbol{\alpha}^\top \boldsymbol{\Sigma} \boldsymbol{\alpha}) + 3\, \Sigma_{kk}\, \beta_j\, (\boldsymbol{\alpha}^\top \boldsymbol{\Sigma} \boldsymbol{\alpha}) + 6\, \beta_j\, \beta_k^2.$$

Fourth–moment term. Similarly,

$$\mathbb{E}[X_j X_k^2\, s] = \sum_m \alpha_m\, \mathbb{E}[X_j X_k^2 X_m] = 2\, \Sigma_{jk}\, \beta_k + \beta_j\, \Sigma_{kk}.$$

Combining,

$$\mathbb{E}[W_j W_k^2] = 6\, \sigma_y^2\, \Sigma_{jk}\, \beta_k + 3\, \sigma_y^2\, \Sigma_{kk}\, \beta_j + 6\, \beta_j\, \beta_k^2.$$

Substituting into (9) with $\gamma_{jk} = \Sigma_{jk} \sigma_y^2 + \beta_j \beta_k$ and $\gamma_{kk} = \Sigma_{kk} \sigma_y^2 + \beta_k^2$, one obtains (8) after straightforward cancellation. □

3 Main Result

Theorem 6 (Variance of the oracle LDSC estimator)**.** *Under the model and notation above, write* $\widetilde{\ell}_j = \ell_j - \mu_\ell$ *and* $\|\widetilde{\boldsymbol{\ell}}\|^2 = \sum_{j=1}^p \widetilde{\ell}_j^2$. *Then*

$$\boxed{\mathrm{Var}(\widetilde{\sigma}_\alpha^2) = \frac{1}{\|\widetilde{\boldsymbol{\ell}}\|^4} \sum_{j=1}^p \sum_{k=1}^p \widetilde{\ell}_j\, \widetilde{\ell}_k\, \mathrm{Cov}\big(\widehat{a}_j^2,\, \widehat{a}_k^2\big),} \tag{10}$$

where, with $\beta_j = (\boldsymbol{\Sigma} \boldsymbol{\alpha})_j$, $\gamma_{jk} = \Sigma_{jk} \sigma_y^2 + \beta_j \beta_k$, $\sigma_y^2 = \boldsymbol{\alpha}^\top \boldsymbol{\Sigma} \boldsymbol{\alpha} + \sigma_\epsilon^2$, *and* ρ_{jkk} *as in* (8),

$$\mathrm{Cov}\big(\widehat{a}_j^2,\, \widehat{a}_k^2\big) = \frac{4\, \beta_j\, \beta_k\, \gamma_{jk}}{n} + \frac{2\, \gamma_{jk}^2 + 2\, \beta_j\, \rho_{jkk} + 2\, \beta_k\, \rho_{kjj}}{n^2} + \frac{\kappa_{jk}}{n^3}, \tag{11}$$

where

$$\kappa_{jk} = \mathbb{E}[U_j^2\, U_k^2] - \gamma_{jj}\, \gamma_{kk} - 2\, \gamma_{jk}^2 \tag{12}$$

is the fourth-order joint cumulant, with $U_j = T_j - \beta_j$.

Proof. Because the denominator $\|\widetilde{\boldsymbol{\ell}}\|^2$ is deterministic (Remark 3), equation (10) is immediate from (3). It remains to establish (11).

Step 1: Reduction to i.i.d. averages. Define $U_{ij} = T_{ij} - \beta_j$ and $\bar{U}_j = n^{-1} \sum_{i=1}^n U_{ij}$, so that $\widehat{a}_j = \beta_j + \bar{U}_j$. Hence

$$\widehat{a}_j^2 - \mathbb{E}[\widehat{a}_j^2] = 2\beta_j\, \bar{U}_j + (\bar{U}_j^2 - n^{-1} \gamma_{jj}).$$

Writing the product $(\widehat{a}_j^2 - \mathbb{E}[\widehat{a}_j^2])(\widehat{a}_k^2 - \mathbb{E}[\widehat{a}_k^2])$ and taking expectations gives

$$\mathrm{Cov}(\widehat{a}_j^2, \widehat{a}_k^2) = 4\beta_j \beta_k\, \mathbb{E}[\bar{U}_j \bar{U}_k] + 2\beta_j\, \mathbb{E}[\bar{U}_j \bar{U}_k^2] + 2\beta_k\, \mathbb{E}[\bar{U}_j^2 \bar{U}_k] + \big(\mathbb{E}[\bar{U}_j^2 \bar{U}_k^2] - n^{-2} \gamma_{jj} \gamma_{kk}\big), \tag{13}$$

using $\mathbb{E}[\bar{U}_j] = 0$.

Step 2: Moments of i.i.d. averages. Since $U_{1j}, \ldots, U_{nj}$ are i.i.d. centered random variables:

(a) $\mathbb{E}[\bar{U}_j \bar{U}_k] = n^{-1}\gamma_{jk}$.

(b) By expanding $\bar{U}_j \bar{U}_k^2 = n^{-3} \sum_{i,i',i''} U_{ij} U_{i'k} U_{i''k}$ and using independence across samples together with $\mathbb{E}[U_{ij}] = 0$, the only nonvanishing contributions arise when $i = i' = i''$; hence $\mathbb{E}[\bar{U}_j \bar{U}_k^2] = n^{-2}\,\rho_{jkk}$.

(c) Expanding $\bar{U}_j^2 \bar{U}_k^2 = n^{-4} \sum_{i_1,i_2,i_3,i_4} U_{i_1 j} U_{i_2 j} U_{i_3 k} U_{i_4 k}$ and partitioning over the three types of nonvanishing index configurations gives

$$\mathbb{E}[\bar{U}_j^2 \bar{U}_k^2] = \frac{1}{n^3}\,\mathbb{E}[U_j^2 U_k^2] + \frac{n-1}{n^3}\big(\gamma_{jj}\gamma_{kk} + 2\gamma_{jk}^2\big).$$

The three surviving partitions of (i_1, i_2, i_3, i_4) are: all four equal (n tuples, contributing $\mathbb{E}[U_j^2 U_k^2]$); $\{i_1 = i_2\} \neq \{i_3 = i_4\}$ ($n(n-1)$ tuples, contributing $\gamma_{jj}\gamma_{kk}$); and $\{i_1 = i_3\} \neq \{i_2 = i_4\}$ or $\{i_1 = i_4\} \neq \{i_2 = i_3\}$ ($2n(n-1)$ tuples, each contributing γ_{jk}^2). Therefore

$$\mathbb{E}[\bar{U}_j^2 \bar{U}_k^2] - n^{-2}\gamma_{jj}\gamma_{kk} = \frac{\kappa_{jk}}{n^3} + \frac{2\,\gamma_{jk}^2}{n^2},$$

where $\kappa_{jk} = \mathbb{E}[U_j^2 U_k^2] - \gamma_{jj}\gamma_{kk} - 2\gamma_{jk}^2$.

Step 3: Assembly. Substituting (a)–(c) into (13):

$$\mathrm{Cov}(\widehat{a}_j^2, \widehat{a}_k^2) = \frac{4\beta_j\beta_k\gamma_{jk}}{n} + \frac{2\beta_j\rho_{jkk} + 2\beta_k\rho_{kjj} + 2\gamma_{jk}^2}{n^2} + \frac{\kappa_{jk}}{n^3}.$$

This is precisely (11). □

4 Explicit Expansion of the Variance

We now substitute the closed-form expressions from Lemmas 4–5 to write each coefficient in (11) purely in terms of the allowed quantities $n, \boldsymbol{\Sigma}, \boldsymbol{\ell}, \boldsymbol{\alpha}, \sigma_\epsilon^2$.

4.1 The $O(n^{-1})$ coefficient

From (7),

$$4\,\beta_j\,\beta_k\,\gamma_{jk} = 4\,\sigma_y^2\,\Sigma_{jk}\,\beta_j\,\beta_k + 4\,\beta_j^2\,\beta_k^2. \tag{14}$$

4.2 The $O(n^{-2})$ coefficient

From (8),

$$2\beta_j\,\rho_{jkk} + 2\beta_k\,\rho_{kjj} = 16\,\sigma_y^2\,\Sigma_{jk}\,\beta_j\,\beta_k + 4\,\sigma_y^2\,(\Sigma_{jj}\beta_k^2 + \Sigma_{kk}\beta_j^2) + 8\,\beta_j^2\beta_k^2.$$

Adding $2\gamma_{jk}^2 = 2\sigma_y^4\Sigma_{jk}^2 + 4\sigma_y^2\Sigma_{jk}\beta_j\beta_k + 2\beta_j^2\beta_k^2$,

$$2\gamma_{jk}^2 + 2\beta_j\rho_{jkk} + 2\beta_k\rho_{kjj} = 2\,\sigma_y^4\,\Sigma_{jk}^2 + 20\,\sigma_y^2\,\Sigma_{jk}\,\beta_j\beta_k + 4\,\sigma_y^2\big(\Sigma_{jj}\beta_k^2 + \Sigma_{kk}\beta_j^2\big) + 10\,\beta_j^2\beta_k^2. \tag{15}$$

4.3 Combined formula

Proposition 7 (Expanded variance)**.** *Define the Hadamard square* $\mathbf{D} = \boldsymbol{\Sigma} \circ \boldsymbol{\Sigma} \in \mathbb{R}^{p\times p}$*, i.e.* $D_{jk} = \Sigma_{jk}^2$*, and write* $\boldsymbol{\beta} = \boldsymbol{\Sigma}\boldsymbol{\alpha}$*,* $\boldsymbol{\beta}^{\circ 2} = (\beta_1^2, \ldots, \beta_p^2)^\top$*,* $\mathbf{d} = (\Sigma_{11}, \ldots, \Sigma_{pp})^\top$*. Then*

$$\begin{aligned}\mathrm{Var}(\widetilde{\sigma}_\alpha^2) = \frac{1}{\|\widetilde{\boldsymbol{\ell}}\|^4}\Bigg\{ & \frac{4\,\sigma_y^2}{n}\,(\widetilde{\boldsymbol{\ell}} \circ \boldsymbol{\beta})^\top \boldsymbol{\Sigma}\,(\widetilde{\boldsymbol{\ell}} \circ \boldsymbol{\beta}) \;+\; \frac{4}{n}\,\big(\widetilde{\boldsymbol{\ell}}^\top \boldsymbol{\beta}^{\circ 2}\big)^2 \\ & +\; \frac{2\,\sigma_y^4}{n^2}\,\widetilde{\boldsymbol{\ell}}^\top \mathbf{D}\,\widetilde{\boldsymbol{\ell}} \;+\; \frac{20\,\sigma_y^2}{n^2}\,(\widetilde{\boldsymbol{\ell}} \circ \boldsymbol{\beta})^\top \boldsymbol{\Sigma}\,(\widetilde{\boldsymbol{\ell}} \circ \boldsymbol{\beta}^{\circ 2}/\boldsymbol{\beta}) \\ & +\; \frac{4\,\sigma_y^2}{n^2}\,\big(\widetilde{\boldsymbol{\ell}}^\top \boldsymbol{\beta}^{\circ 2}\big)\big(\widetilde{\boldsymbol{\ell}}^\top \mathbf{d}\big) \;+\; \frac{4\,\sigma_y^2}{n^2}\,\big(\widetilde{\boldsymbol{\ell}}^\top \mathbf{d}\big)\big(\widetilde{\boldsymbol{\ell}}^\top \boldsymbol{\beta}^{\circ 2}\big) \\ & +\; \frac{10}{n^2}\,\big(\widetilde{\boldsymbol{\ell}}^\top \boldsymbol{\beta}^{\circ 2}\big)^2 \;+\; \frac{1}{n^3}\sum_{j,k} \widetilde{\ell}_j\,\widetilde{\ell}_k\,\kappa_{jk}\Bigg\}. \end{aligned} \tag{16}$$

Because the middle lines above contain a notational ambiguity when $\beta_j = 0$*, we restate the formula more carefully using index summations. Substituting* (14)–(15) *into* (10)–(11)*:*

$$\boxed{\begin{aligned}\mathrm{Var}(\widetilde{\sigma}_\alpha^2) = \frac{1}{\|\widetilde{\boldsymbol{\ell}}\|^4}\Bigg\{ & \frac{4\sigma_y^2}{n}\sum_{j,k} \widetilde{\ell}_j\widetilde{\ell}_k\,\Sigma_{jk}\beta_j\beta_k \;+\; \frac{4}{n}\Big(\sum_j \widetilde{\ell}_j\beta_j^2\Big)^2 \\ & +\; \frac{2\sigma_y^4}{n^2}\sum_{j,k} \widetilde{\ell}_j\widetilde{\ell}_k\,\Sigma_{jk}^2 \;+\; \frac{20\sigma_y^2}{n^2}\sum_{j,k} \widetilde{\ell}_j\widetilde{\ell}_k\,\Sigma_{jk}\beta_j\beta_k \\ & +\; \frac{4\sigma_y^2}{n^2}\sum_{j,k} \widetilde{\ell}_j\widetilde{\ell}_k(\Sigma_{jj}\beta_k^2 + \Sigma_{kk}\beta_j^2) \;+\; \frac{10}{n^2}\Big(\sum_j \widetilde{\ell}_j\beta_j^2\Big)^2 \\ & +\; \frac{1}{n^3}\sum_{j,k} \widetilde{\ell}_j\widetilde{\ell}_k\,\kappa_{jk}\Bigg\}\end{aligned}} \tag{17}$$

where $\sigma_y^2 = \boldsymbol{\alpha}^\top\boldsymbol{\Sigma}\boldsymbol{\alpha} + \sigma_\epsilon^2$*,* $\beta_j = (\boldsymbol{\Sigma}\boldsymbol{\alpha})_j$*, and the fourth-order cumulant* κ_{jk} *is defined in* (12)*.*

Equivalently, using the compact notation $\mathbf{D} = \boldsymbol{\Sigma} \circ \boldsymbol{\Sigma}$ and $\mathbf{d} = \mathrm{diag}(\boldsymbol{\Sigma})$, the $O(n^{-2})$ factor $\sum_{j,k} \widetilde{\ell}_j\widetilde{\ell}_k(\Sigma_{jj}\beta_k^2 + \Sigma_{kk}\beta_j^2) = 2(\widetilde{\boldsymbol{\ell}}^\top\mathbf{d})(\widetilde{\boldsymbol{\ell}}^\top\boldsymbol{\beta}^{\circ 2})$.

Remark 8 (Interpretation of scale). Each line of (17) has a clear statistical origin:

- The $O(n^{-1})$ terms arise from the interaction between the signal $\boldsymbol{\beta} = \boldsymbol{\Sigma}\boldsymbol{\alpha}$ and both the genetic sampling noise and the environmental noise σ_ϵ^2. They vanish when $\boldsymbol{\alpha} = \mathbf{0}$.
- The $O(n^{-2})$ terms arise from (i) the purely noise-driven variance $2\sigma_y^4\Sigma_{jk}^2/n^2$, (ii) signal–noise cross-terms involving third moments, and (iii) squared-signal contributions.
- The $O(n^{-3})$ term κ_{jk} captures residual non-Gaussianity of the quadratic forms in $\widehat{a}_j^2$; it is computable via Isserlis' theorem (eighth moments of the joint Gaussian $(\mathbf{x}, \epsilon)$) but is typically negligible for moderately large n.

Remark 9 (Null case $\boldsymbol{\alpha} = \mathbf{0}$). When all genetic effects are zero, $\beta_j = 0$ for all j and $\sigma_y^2 = \sigma_\epsilon^2$. The variance simplifies to

$$\mathrm{Var}(\widetilde{\sigma}_\alpha^2)\Big|_{\boldsymbol{\alpha}=\mathbf{0}} = \frac{2\,\sigma_\epsilon^4}{n^2} \cdot \frac{\widetilde{\boldsymbol{\ell}}^\top \mathbf{D}\, \widetilde{\boldsymbol{\ell}}}{\|\widetilde{\boldsymbol{\ell}}\|^4} + O(n^{-3}),$$

recovering the classical variance of a Haseman–Elston–type regression under the null.

Remark 10 (Role of the block-diagonal structure). The assumption that $\boldsymbol{\Sigma}$ is block-diagonal is used in two ways. First, it ensures that $\ell_j = \sum_k \Sigma_{jk}^2$ and all the sums involving Σ_{jk} are effectively finite-range (supported within blocks), so that all moments $\mathbb{E}[U_j^a U_k^b]$ for j, k in different blocks factor into independent contributions. Second, it guarantees that the fourth-order cumulant κ_{jk} in (12) vanishes whenever j and k lie in different LD blocks, since in that case U_j and U_k are independent up to the shared noise ϵ, and the dependence through ϵ alone contributes only through lower-order terms already accounted for. Thus the $O(n^{-3})$ sum in (17) is a within-block sum.